

The *plain flap* shown in figure 1.17 is a simple hinged portion of the trailing edge. The effect of the camber added well aft on the chord causes a significant increase in $c_{l_{max}}$. In addition, the zero lift angle changes to a more negative value and the drag increases greatly. The *split flap* shown in figure 1.17 consist of plate deflected from the lower surface of the section and produces a slightly greater change in $c_{l_{max}}$ than the plain flap. However, a much larger change in drag results from the great turbulent wake produced by this type flap. The greater drag may not be such a disadvantage when it is realized that it may be advantageous to accomplish steeper landing approaches over obstacles or require higher power from the engine during approach (to minimize engine acceleration time for waveoff).

The *slotted flap* is similar to the plain flap but the gap between the main section and flap leading edge is given specific contours. High energy air from the lower surface is ducted to the flap upper surface. The high energy air from the slot accelerates the upper surface boundary layer and delays airflow separation to some higher lift coefficient. The slotted flap can cause much greater increases in $c_{l_{max}}$ than the plain or split flap and section drags are much lower.

The *Fowler flap* arrangement is similar to the slotted flap. The difference is that the deflected flap segment is moved aft along a set of tracks which increases the chord and effects an increase in wing area. The Fowler flap is characterized by large increases in $c_{l_{max}}$ with minimum changes in drag.

One additional factor requiring consideration in a comparison of flap types is the aerodynamic twisting moments caused by the flap. Positive camber produces a nose down twisting moment—especially great when large camber is used well aft on the chord (an obvious implication is that flaps are not practical on a flying wing or tailless airplane). The deflection of a flap causes large nose down moments which create important twisting

loads on the structure and pitching moments that must be controlled with the horizontal tail. Unfortunately, the flap types producing the greatest increases in $c_{l_{max}}$ usually cause the greatest twisting moments. The Fowler flap causes the greatest change in twisting moment while the split flap causes the least. This factor—along with mechanical complexity of the installation—may complicate the choice of a flap configuration.

The effectiveness of flaps on a wing configuration depend on many different factors. One important factor is the amount of the wing area affected by the flaps. Since a certain amount of the span is reserved for ailerons, the actual wing maximum lift properties will be less than that of the flapped two-dimensional section. If the basic wing has a low thickness, any type of flap will be less effective than on a wing of greater thickness. Sweepback of the wing can cause an additional significant reduction in the effectiveness of flaps.

High lift devices applied to the leading edge of a section consist of slots, slats, and small amounts of local camber. The fixed slot in a wing conducts flow of high energy air into the boundary layer on the upper surface and delays airflow separation to some higher angle of attack and lift coefficient. Since the slot alone effects no change in camber, the higher maximum lift coefficient will be obtained at a higher angle of attack, i.e., the slot simply delays stall to a higher angle of attack. An automatic slot arrangement consists of a leading edge segment (slat) which is free to move on tracks. At low angles of attack the slat is held flush against the leading edge by the high positive local pressures. When the section is at high angles of attack, the high local suction pressures at the leading edge create a chordwise force forward to actuate the slat. The slot formed then allows the section to continue to a higher angle of attack and produce a $c_{l_{max}}$ greater than that of the